

;***** main.s *****

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; Date Created: 2/3/26

; Last Modified:

; Brief description of the program:

;

THUMB

AStudent1	EQU	3
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AStudent2	EQU	2
-----------	-----	---

AStudent3	EQU	1
-----------	-----	---

AStudent4	EQU	1
-----------	-----	---

AStudent5	EQU	3
-----------	-----	---

BStudent1	EQU	5
-----------	-----	---

BStudent2	EQU	1
-----------	-----	---

BStudent3	EQU	4
-----------	-----	---

BStudent4	EQU	3
-----------	-----	---

BStudent5	EQU	2
-----------	-----	---

CStudent1	EQU	7
-----------	-----	---

CStudent2	EQU	4
-----------	-----	---

CStudent3	EQU	5
-----------	-----	---

CStudent4 EQU 3

CStudent5 EQU 6

GroupN EQU 5

; *****

; ***** CODE AREA IN ROM *****

; *****

; ROM addresses start at address 0x0000.0000, but your code starts a little after that

AREA |.text|, CODE, READONLY, ALIGN=2 ; Flash EEPROM

EXPORT Start

; * Program begins here!

Start

; ***** TASK 1 *****

; *****

; Means

; R12 - MEANS FOR GROUP A, R11 = MEANS FOR GROUP B, R10 = MEANS FOR GROUP C

; R9 = GRAND MEAN <- THIS IS THE AVERAGE MEAN FOR ALL OF THE GROUPS.

; GROUP A

;Load the resistors values for group A

MOV R0, #AStudent1

MOV R1, #AStudent2

MOV R2, #AStudent3

MOV R3, #AStudent4

MOV R4, #AStudent5

; Summ of Group A

ADD R5, R0,R1

ADD R5, R5,R2

ADD R5, R5,R3

ADD R5, R5,R4

; THIS WILL GATHER THE NUMBER OF GROUPS

MOV R6,#GroupN

SDIV R12,R5,R6 ; THIS LINE WILL GIVE US THE MEAN OF GROUP A

;-----

; GROUP B

;Load the resistors values for group B

MOV R0, #BStudent1

MOV R1, #BStudent2

MOV R2, #BStudent3

MOV R3, #BStudent4

MOV R4, #BStudent5

; Summ of Group B

ADD R5, R0,R1

ADD R5, R5,R2

ADD R5, R5,R3

ADD R5, R5,R4

SDIV R11,R5,R6 ; THIS LINE WILL GIVE US THE MEAN OF GROUP B

;-----

; GROUP C

;Load the resistors values for group C

MOV R0, #CStudent1

MOV R1, #CStudent2

MOV R2, #CStudent3

MOV R3, #CStudent4

MOV R4, #CStudent5

; Summ of Group C

ADD R5, R0,R1

ADD R5, R5,R2

ADD R5, R5,R3

ADD R5, R5,R4

SDIV R10,R5,R6 ; THIS LINE WILL GIVE US THE MEAN OF GROUP B

;-----

; THE GRAND MEAN

ADD R0, R12, R11

ADD R0, R0, R10

MOV R1, #3 ; THIS WILL ALLOW US TO DIVIDE BY 3 GROUPS

SDIV R9, R0, R1 ; GRAND MEAN IS IN R9

; ***** TASK 2 *****

; *****

; Sum of Squared Errors (SS) by Group

; WE NEED TO FIND THE SUM OF SQUARED ERRORS BY GROUPS

; R8 IS FOR GROUP A

; R7 IS FOR GROUP B

; R6 IS FOR GROUP C

;-----

; GROUP A

MOV R8, #0

; UNDERSTAND THAT WE NEED TO CALCULATE (ASTUDENT1 - MEAN_A)^2

MOV R0, #ASTUDENT1

SUB R1, R0, R12 ; THIS IS X-MEAN

MUL R1, R1, R1 ; THIS MAKE (X-MEAN)^2

ADD R8, R8, R1

; CONTINUE DOING THIS FOR ALL OF THE STUDENTS IN GROUP A, B, C

; STUDENT 2

```

MOV R0, #AStudent2

SUB R1,R0,R12 ; THIS IS X-MEAN

MUL R1,R1,R1 ; THIS MAKE (X-MEAN)^2

ADD R8,R8,R1

;STUDENT 3

MOV R0, #AStudent3

SUB R1,R0,R12 ; THIS IS X-MEAN

MUL R1,R1,R1 ; THIS MAKE (X-MEAN)^2

ADD R8,R8,R1

;STUDENT 4

MOV R0, #AStudent4

SUB R1,R0,R12 ; THIS IS X-MEAN

MUL R1,R1,R1 ; THIS MAKE (X-MEAN)^2

ADD R8,R8,R1

;STUDENT 5

MOV R0, #AStudent5

SUB R1,R0,R12 ; THIS IS X-MEAN

MUL R1,R1,R1 ; THIS MAKE (X-MEAN)^2

ADD R8,R8,R1 ; NOW R8 SHOULD HAVE ALL OF THE SSE FOR GROUP A.

;-----

;GROUP B

MOV R7,#0

;STUDENT 1

MOV R0, #BStudent1

SUB R1,R0,R11 ; THIS IS X-MEAN

MUL R1,R1,R1 ; THIS MAKE (X-MEAN)^2

```

```

ADD R7,R7,R1
;STUDENT 2
MOV R0, #BStudent2
SUB R1,R0,R11 ; THIS IS X-MEAN
MUL R1,R1,R1 ; THIS MAKE (X-MEAN)^2
ADD R7,R7,R1
;STUDENT 3
MOV R0, #BStudent3
SUB R1,R0,R11 ; THIS IS X-MEAN
MUL R1,R1,R1 ; THIS MAKE (X-MEAN)^2
ADD R7,R7,R1
;STUDENT 4
MOV R0, #BStudent4
SUB R1,R0,R11 ; THIS IS X-MEAN
MUL R1,R1,R1 ; THIS MAKE (X-MEAN)^2
ADD R7,R7,R1
;STUDENT 5
MOV R0, #BStudent5
SUB R1,R0,R11 ; THIS IS X-MEAN
MUL R1,R1,R1 ; THIS MAKE (X-MEAN)^2
ADD R7,R7,R1
;-----
;GROUP C
MOV R6,#0
;STUDENT 1
MOV R0, #CStudent1

```

```
SUB R1,R0,R10 ; THIS IS X-MEAN
MUL R1,R1,R1 ; THIS MAKE (X-MEAN)^2
ADD R6,R6,R1
;STUDENT 2
MOV R0, #CStudent2
SUB R1,R0,R10 ; THIS IS X-MEAN
MUL R1,R1,R1 ; THIS MAKE (X-MEAN)^2
ADD R6,R6,R1
;STUDENT 3
MOV R0, #CStudent3
SUB R1,R0,R10 ; THIS IS X-MEAN
MUL R1,R1,R1 ; THIS MAKE (X-MEAN)^2
ADD R6,R6,R1
;STUDENT 4
MOV R0, #CStudent4
SUB R1,R0,R10 ; THIS IS X-MEAN
MUL R1,R1,R1 ; THIS MAKE (X-MEAN)^2
ADD R6,R6,R1
;STUDENT 5
MOV R0, #CStudent5
SUB R1,R0,R10 ; THIS IS X-MEAN
MUL R1,R1,R1 ; THIS MAKE (X-MEAN)^2
ADD R6,R6,R1
```

; NEED TO DOUBLE CHECK TASK 2 !!!!!!!!!!!


```
; ***** TASK 3 *****
```

```
; *****
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```
; SSR -> Regression Sum of Squares
```

```
; R5 = (meanA - grandMean)^2 + (meanB - grandMean)^2 + (meanC - grandMean)^2
```

```
MOV R5, #0
```

```
; (R12 - R9)^2
```

```
SUB R0, R12, R9
```

```
MUL R0, R0, R0
```

```
ADD R5, R5, R0
```

```
; (R11 - R9)^2
```

```
SUB R0, R11, R9
```

```
MUL R0, R0, R0
```

```
ADD R5, R5, R0
```

```
; (R10 - R9)^2
```

```
SUB R0, R10, R9
```

```
MUL R0, R0, R0
```

```
ADD R5, R5, R0
```

```
; ***** TASK 4 *****
```

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; *****
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```
; SSE -> error sum of squares total
```

```
; SSE (R4) = SS1 (R8) + SS2 (R7) + SS3 (R6)
```

```
ADD R4, R8, R7
```

```
ADD R4, R4, R6
```

```
ADD R3, R5, R4
```

```
NOP
```

```
ALIGN ; make sure the end of this section is aligned
```

```
END ; end of file
```